Yassa Ashraf

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineer with strong experience in backend, full-stack, and cloud-native systems. Proven ability to design, build, and scale production services using Java, TypeScript, Node.js, NestJS, ASP.NET, and AWS. Experienced in microservices, clean architecture, CI/CD, and observability. Delivered customer-facing features across startup and enterprise environments with measurable performance and reliability improvements.

EDUCATION

Bachelor of Computer Science (Software Engineering) — German International University, Cairo
Oct 2021 – Jul 2025

BACHELOR THESIS

Using Explainable AI Algorithms for Material Recognition Grade: A+

- Built ChemBERTa-based material recognition models with post-hoc explainability techniques.
- Enabled traceable predictions using attention visualization and feature attribution methods.
- Deployed inference services using Flask on AWS (SageMaker, EC2, S3) with CI/CD pipelines.
- Delivered a Next.js dashboard for real-time inspection of model outputs and explanations.

PUBLICATIONS

MoleculeX-Predictive: An augmentation of transformers with explainable AI and knowledge layers for efficient drug property prediction
Yassa Ashraf, Daniel Sameh, John Zaki, Amal Aboulhassan
Array (Elsevier), 2026
DOI: 10.1016/j.array.2026.100820
Link: ScienceDirect

CERTIFICATIONS

Oracle Cloud Infrastructure Foundations Associate (2025) Issued Mar 2025
Credential Verification

- **Career Essentials in Generative AI** — Microsoft / LinkedIn Issued Oct 2025
- **Ethics in the Age of Generative AI** — LinkedIn Issued Oct 2025
- **Explainable AI (XAI)** — Duke University Issued Oct 2025
- **Interpretable Machine Learning** — Duke University Issued Oct 2025

TECHNICAL SKILLS

Programming Languages: Java, TypeScript, JavaScript, Python, SQL, C++, C#, Dart, Bash, Kotlin, Go
Backend Technologies: Spring Boot, Spring MVC, Microservices, J2EE, NestJS, Node.js, Express, ASP.NET Web API, REST, GraphQL
Frontend Technologies: React, Next.js, Angular, React Native, Tailwind CSS
Databases and Data Stores: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis
ORM and Data Access: Prisma, Drizzle ORM, JDBC
Cloud and DevOps: AWS (EC2, Lambda, ECS, EKS, S3, SageMaker, API Gateway, CloudFront, CloudWatch), OCI, Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Kong API Gateway
Architecture and Practices: Clean Architecture, Domain-Driven Design, Microservices, Event-Driven Systems, OOP, Design Patterns
API Management and Gateway: AWS API Gateway, Kong Gateway, API Design, Rate Limiting, Authentication (JWT, OAuth2), Request Routing, API Versioning
Security: JWT, OAuth2, Role-Based Access Control (RBAC), API Security, API Gateway Authentication, Rate Limiting, Throttling

Testing and Quality: JUnit, Jest, Unit Testing, Integration Testing
Monitoring and Observability: CloudWatch, Grafana, Logging, Metrics
Tools: Git, Gradle, pnpm, npm, Postman, Linux, Bash, Nginx

PROFESSIONAL EXPERIENCE

Associate Software Engineer — Vodafone Intelligent Solutions (VOIS) Cairo, Egypt
Jul 2025 – Present

- Developed and enhanced enterprise ASP.NET Web API services in an N-tier architecture.
- Implemented advanced filtering, pagination, and reusable API components.
- Refactored Angular modules to improve maintainability and reduce duplicated logic.
- Improved error handling, logging, and validation to increase system reliability.
- Supported UAT cycles and stakeholder demos while aligning solutions with security standards.

Intern Software Engineer — Vodafone Intelligent Solutions (VOIS) Cairo, Egypt
Feb 2025 – Jul 2025

- Contributed to backend Web API enhancements and Angular UI refinements.
- Improved p95 response times for critical endpoints through query optimization.
- Implemented unit tests and participated in peer code reviews.

AI Software Engineer — VIXI AI Remote, UAE
Sep 2025 – Jan 2026

- Designed and implemented NLP pipelines using transformer-based models and retrieval techniques.
- Built containerized inference services with autoscaling and monitoring.
- Collaborated with product and data teams to prioritize enterprise features.
- Reduced deployment and iteration cycles by introducing preview and staging environments.

Software Engineer — Xylem Software Remote, Egypt
Jun 2025 – Nov 2025

- Delivered web and mobile features in a monorepo using React, React Native, and Node.js.
- Led backend development for analytics platforms including ingestion APIs and dashboards.
- Improved application performance via caching strategies and database query optimization.

Mid-Level Backend Engineer — Trevi Cairo, Egypt
Oct 2024 – Sep 2025

- Architected wallet and loyalty microservices using NestJS and Prisma.
- Integrated cashback and promotion engines serving high transaction volumes.
- Deployed services on AWS with health checks, retries, and SLA monitoring.

Software Engineer — MM Solutions Remote, USA
Jul 2024 – Mar 2025

- Led web integration initiatives and optimized frontend bundle size and caching.
- Automated dispatch workflows with Amazon Freight partners.

Full Stack Developer — Kinetics Egypt Cairo, Egypt
Aug 2024 – Oct 2024

- Delivered AWS-based solutions using EC2 and S3 with secure configurations.
- Built features in collaboration with product teams under tight deadlines.

Full Stack Developer (Freelance) — Fiverr Remote
May 2023 – Oct 2023

- Built custom web applications, dashboards, and games for international clients.
- Managed full delivery lifecycle from requirements to deployment.

Full Stack Developer Intern — Instatus Cairo, Egypt
Jul 2022 – Sep 2022

- Developed frontend features using Next.js, GraphQL, and Tailwind CSS.
- Integrated GraphQL queries and mutations to support real-time status updates.
- Improved UI consistency and responsiveness across dashboards.
- Collaborated with backend engineers to align schema design with frontend needs.

SELECTED PROJECTS

Beity App — Cloud Kitchens Ordering and Delivery System NestJS, React Native, AWS, PostgreSQL

- Built a full-stack platform for ordering from cloud kitchens with delivery management.
- Implemented order lifecycle, role-based access, and Clean Architecture backend.
- Deployed on AWS with CI/CD and secure environment configuration.

Intervyo — Smart Interview Platform Next.js, Node.js, TypeScript, PostgreSQL

- Developed an interview management platform for technical and HR workflows.
- Implemented scheduling, evaluations, and REST APIs with scalable architecture.

ECS Help Desk System MERN Stack, Tailwind CSS

- Built a ticketing system with roles, SLAs, real-time updates, and analytics.
- Improved agent productivity through optimized workflows.

E-Commerce Platform for Plastic Pallets NestJS, Next.js, Kafka

- Designed microservices for catalog, orders, and payments.
- Used Kafka for asynchronous communication and resilience.

Amaria E-Commerce — amaria-local.vercel.app Next.js, TypeScript

- Built an online store for apparel with secure checkout and delivery flow.
- Focused on responsive UI and performance.

Myko E-Commerce — myko-store.vercel.app Next.js, TypeScript

- Developed a scalable apparel e-commerce platform with cart and payments.
- Emphasized usability and clean frontend architecture.

Cairo Metro Reservation System Node.js, Express, Bootstrap

- Built booking APIs with fare calculation and session handling.
- Developed responsive interfaces for reservations.

Task Tian — Task Management Platform GraphQL, Prisma, Next.js

- Implemented task prioritization, filters, and role-based GraphQL APIs.

Bookstore Serverless Application AWS Lambda, API Gateway, DynamoDB

- Built serverless CRUD and search APIs using AWS API Gateway and Lambda.
- Implemented request routing, authentication, and rate limiting at API Gateway layer.
- Integrated DynamoDB with scalable and low-latency access patterns.
- Added monitoring and logging using CloudWatch with CI/CD pipelines.

Bachelor Portal System Flutter, SQL

- Implemented transactional student workflows and optimized queries.

LANGUAGES

English: Professional Proficiency
Arabic: Native Proficiency
German: Limited Working Proficiency